Visualization of Network Neurons



Network Stimulation (for class c)

- Natural Input Stimuli Retrieval
  - simple search over the dataset and probably some natural augmentations of it

$$S_{natural} = \max_{\mathbf{x} \in \mathcal{D}} P(\mathrm{y} = \mathrm{c}|\mathbf{x})$$

- Activation Maximization (construct “optimal“ stimuli)
  - complicated search over the input space (usually done with constraints for the optimization)

$$S_{artificial} = \max_{\mathbf{x} \in \mathbb{R}^d} P(\mathrm{y} = \mathrm{c}|\mathbf{x})$$

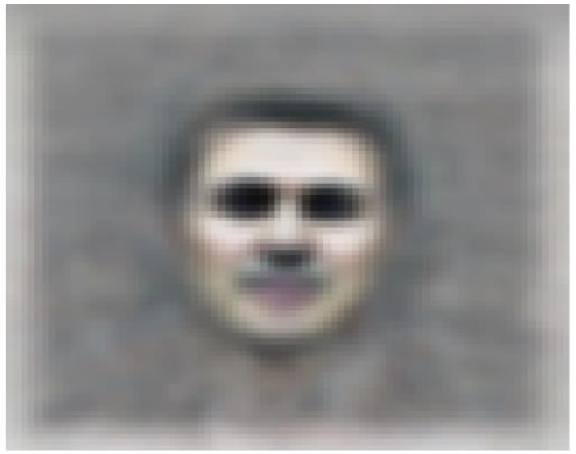


Model Visualization

Network Stimulation

- Unsupervised Sparse AutoEncoder on YouTube
- 1000 machines (16,000 cores)





numerical optimization to find the optimal stimulus (e.g., find a norm bound input that maximizes a certain neuron activity)

visualizing the most responsive stimuli in the test set.

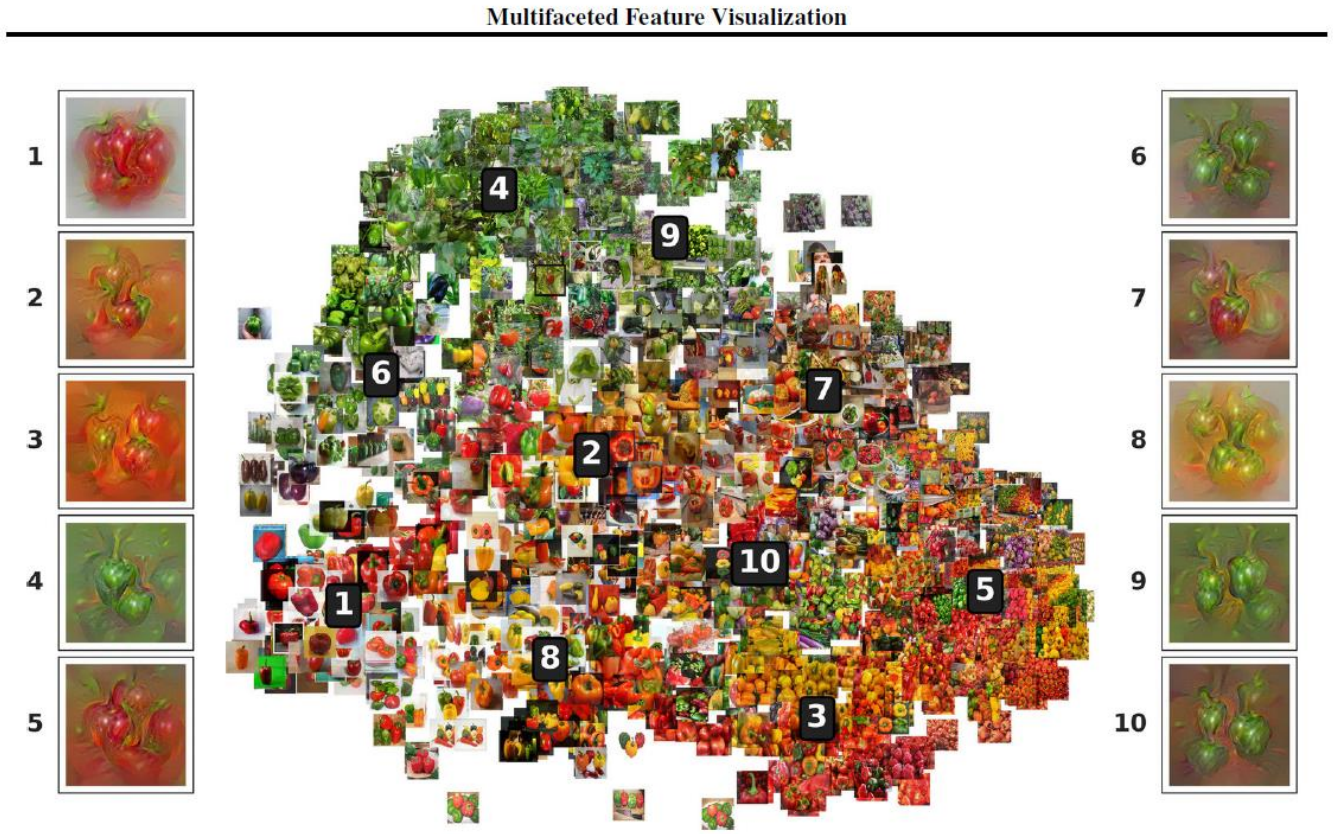


https://www.wired.com/2012/06/google-x-neural-network/

[Le et al. “Building High-level Features Using Large Scale Unsupervised Learning”, ICML 2012]

Network Stimulation
Multi-model activations

- clustering of the inputs
- average each cluster into a centroid
- initialize activation maximization with each centroid separately



["Deep Dream": https://deepdreamgenerator.com/]

[Nguyan et al. "Multifaceted Feature Visualization: Uncovering the Different Types of Features Learned By Each Neuron in Deep Neural Networks", 2017]

Model Visualization

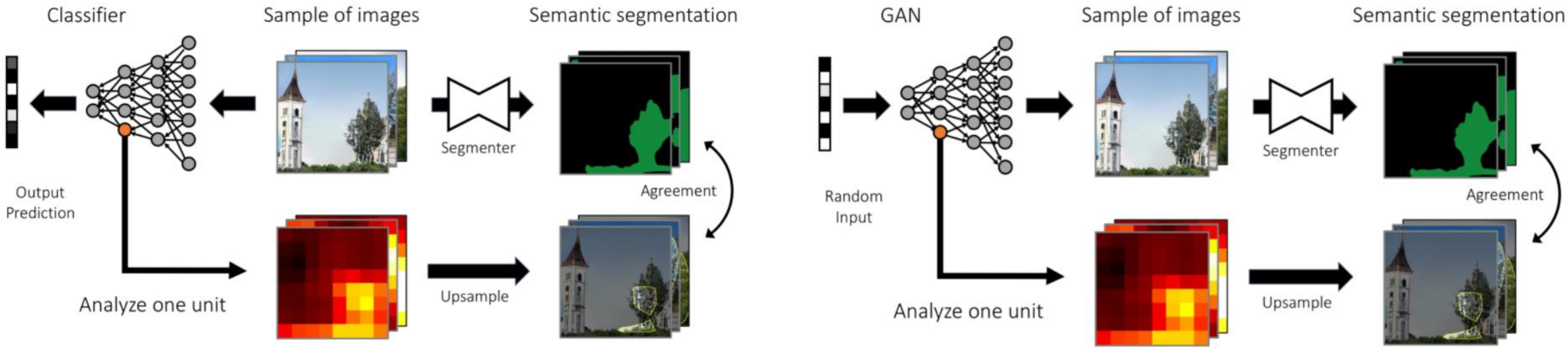
- **Network Stimulation**
  Multi-model activations



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GAN Dissection



[https://gandissect.csail.mit.edu/]

[Bau et al. "Understanding the Role of Individual Units in a Deep Network", PNAS 2020]

Model Visualization

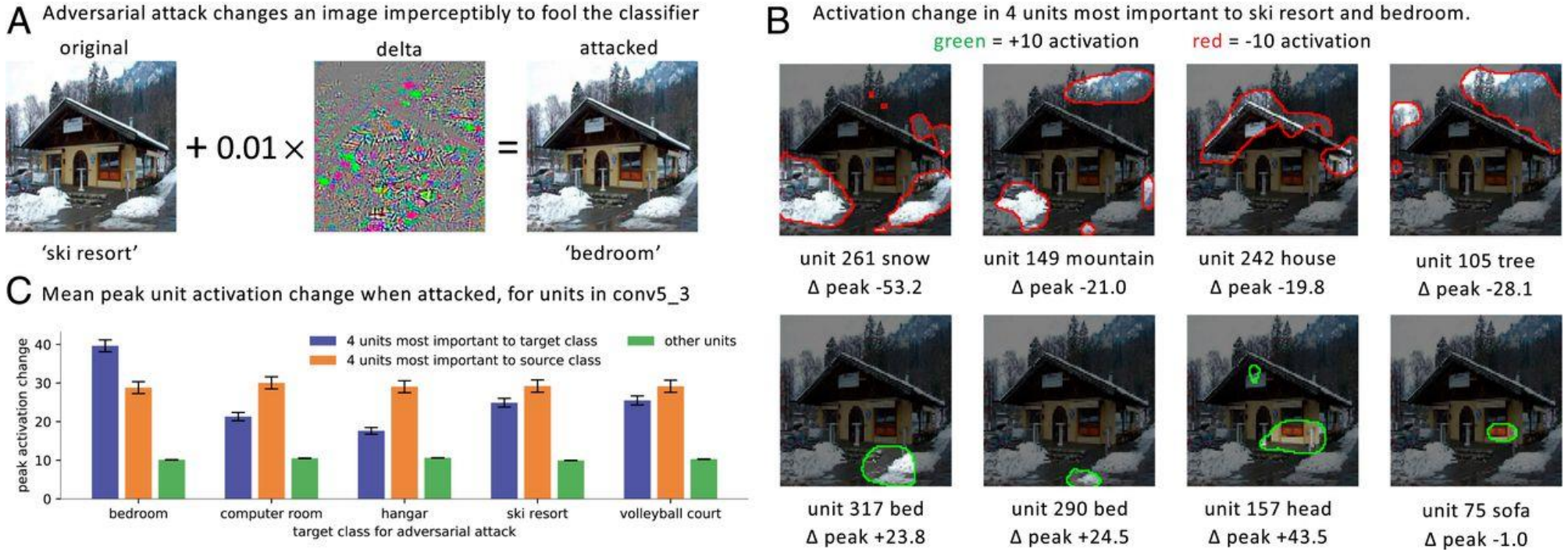
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